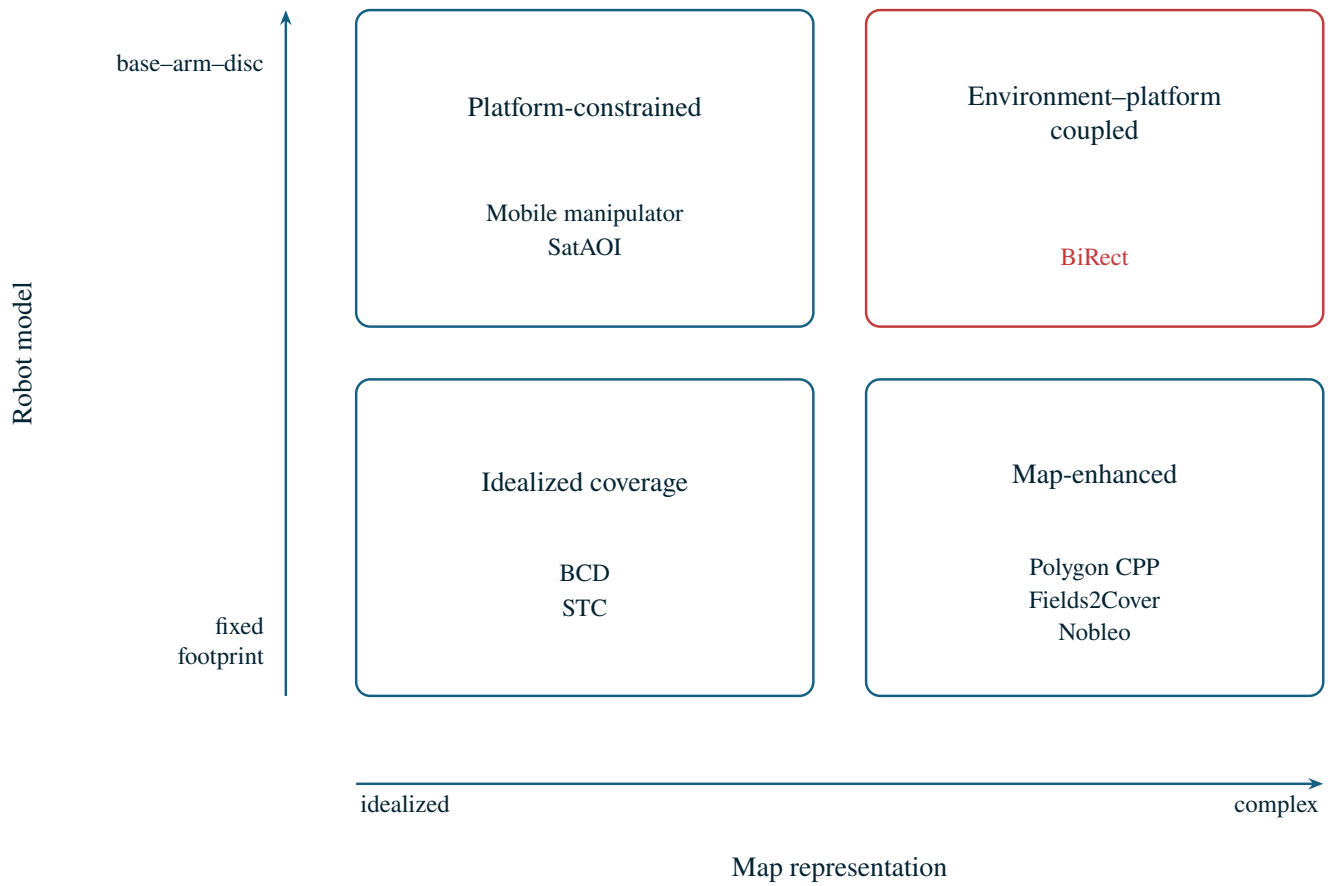
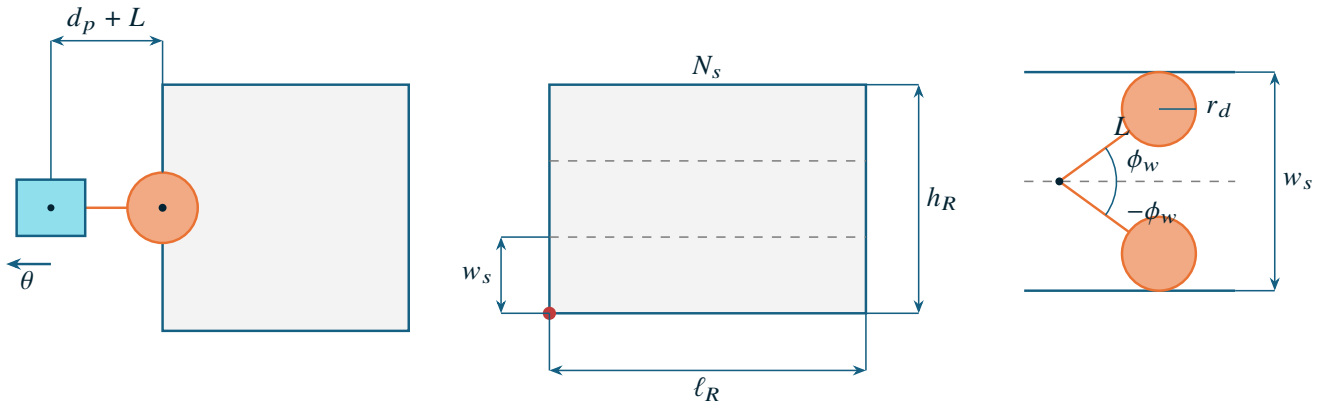
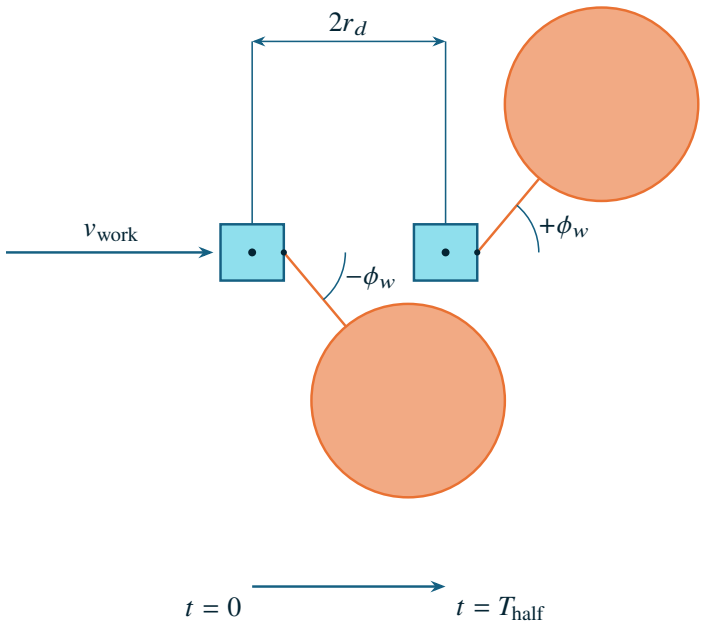
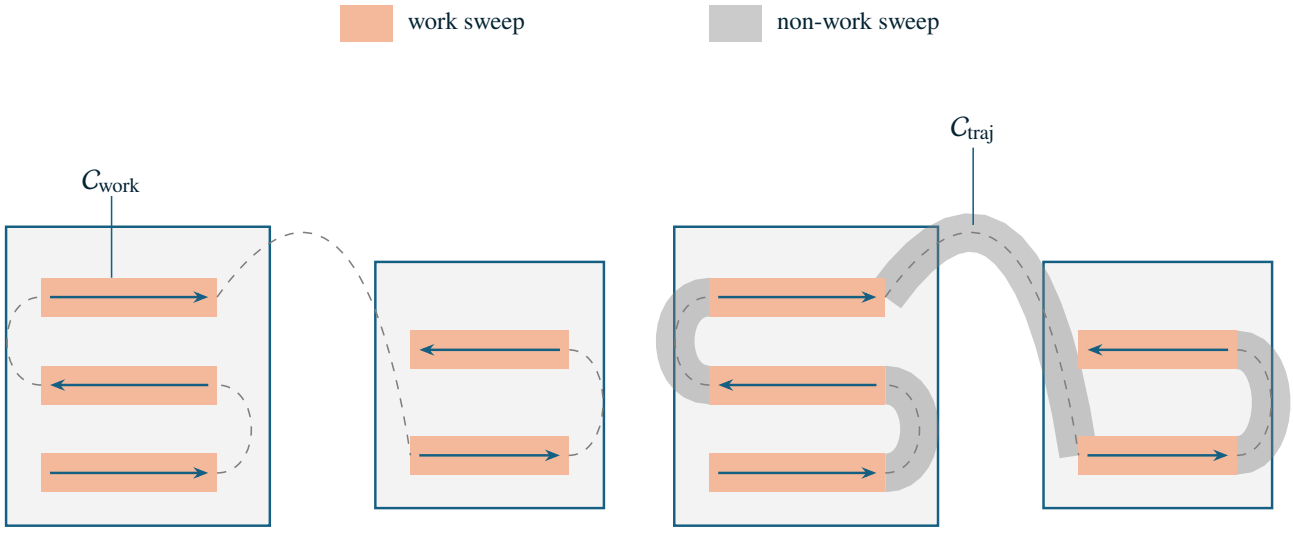
Redrawn figures



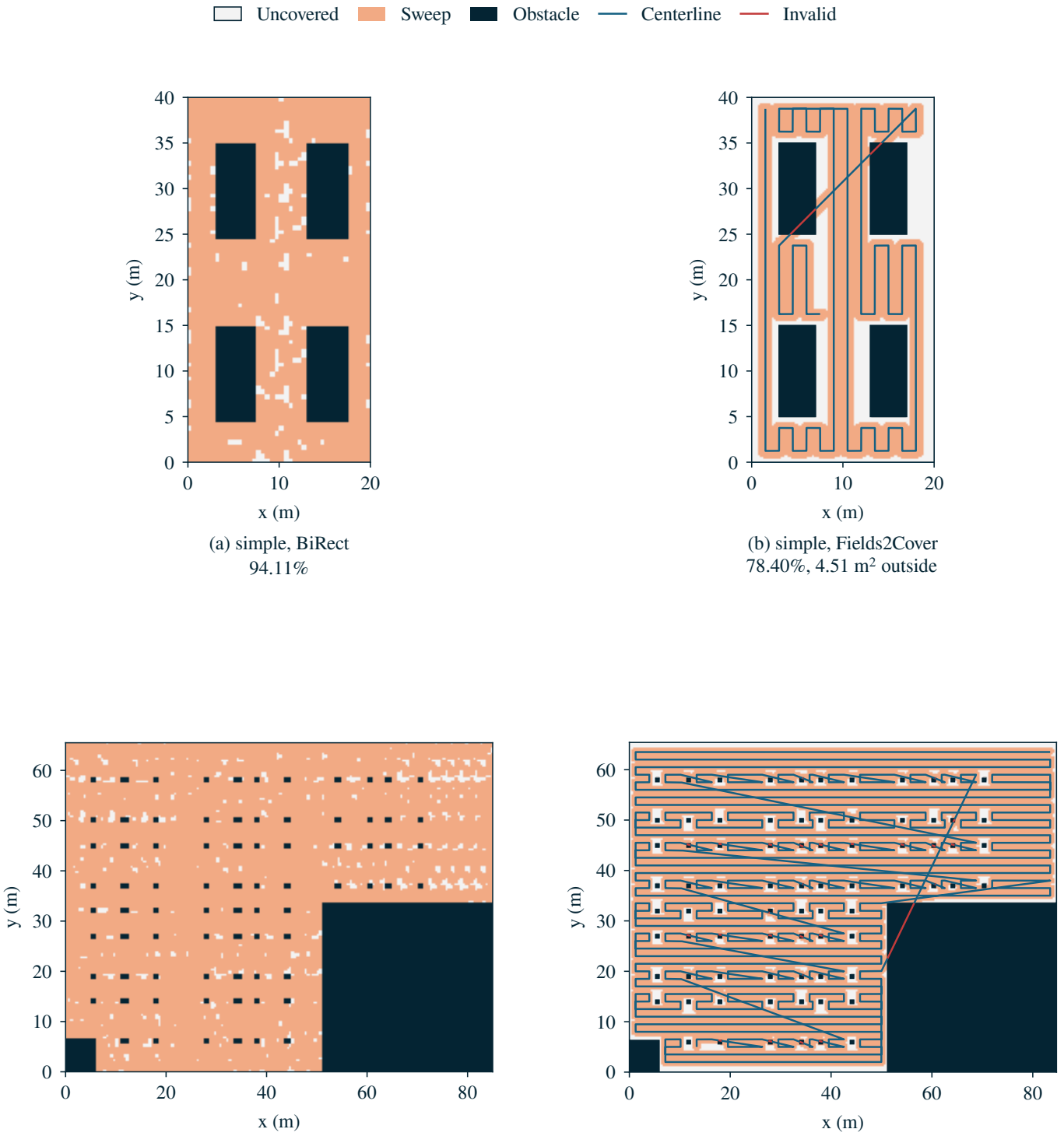


(a) (b) (c)


$2r_d$
v_{work}
$+\phi_w$
$-\phi_w$
$t = 0$
$t = T_{\text{half}}$


work sweep
non-work sweep
C_{work}
C_{traj}

(a) (b)

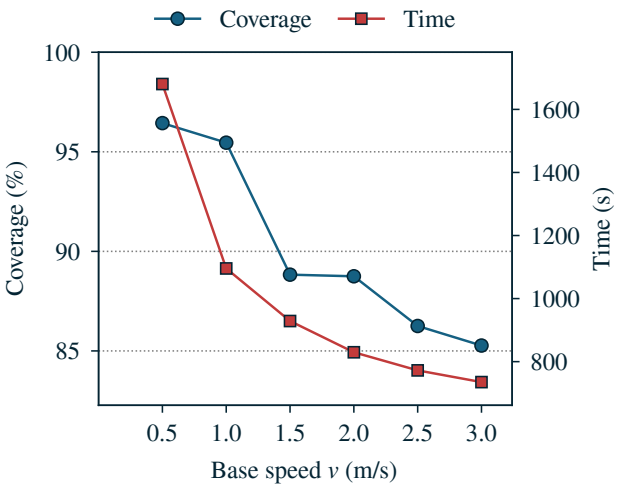


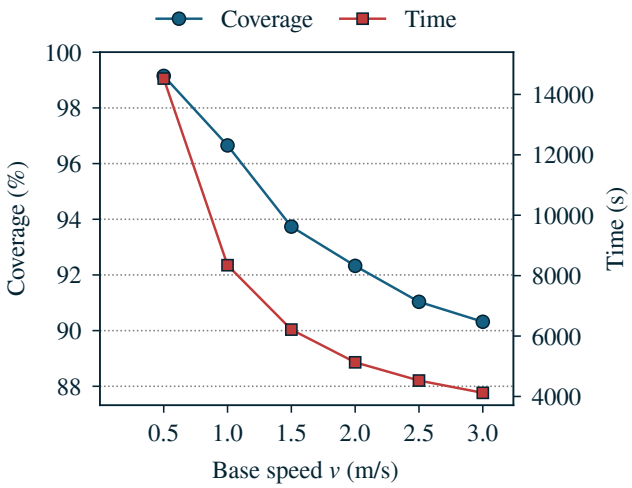
(a) simple, BiRect
94.11%

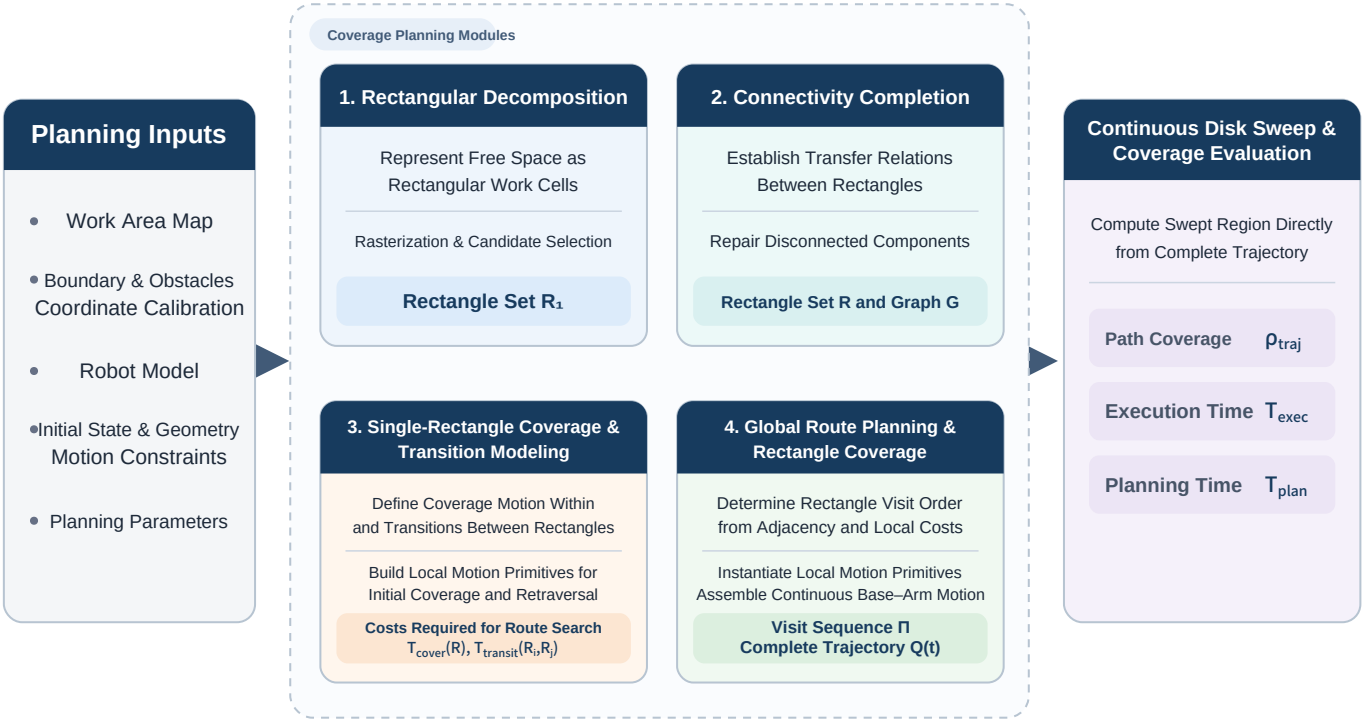
(b) simple, Fields2Cover
78.40%, 4.51 m^2 outside

(c) complex, BiRect
95.73%

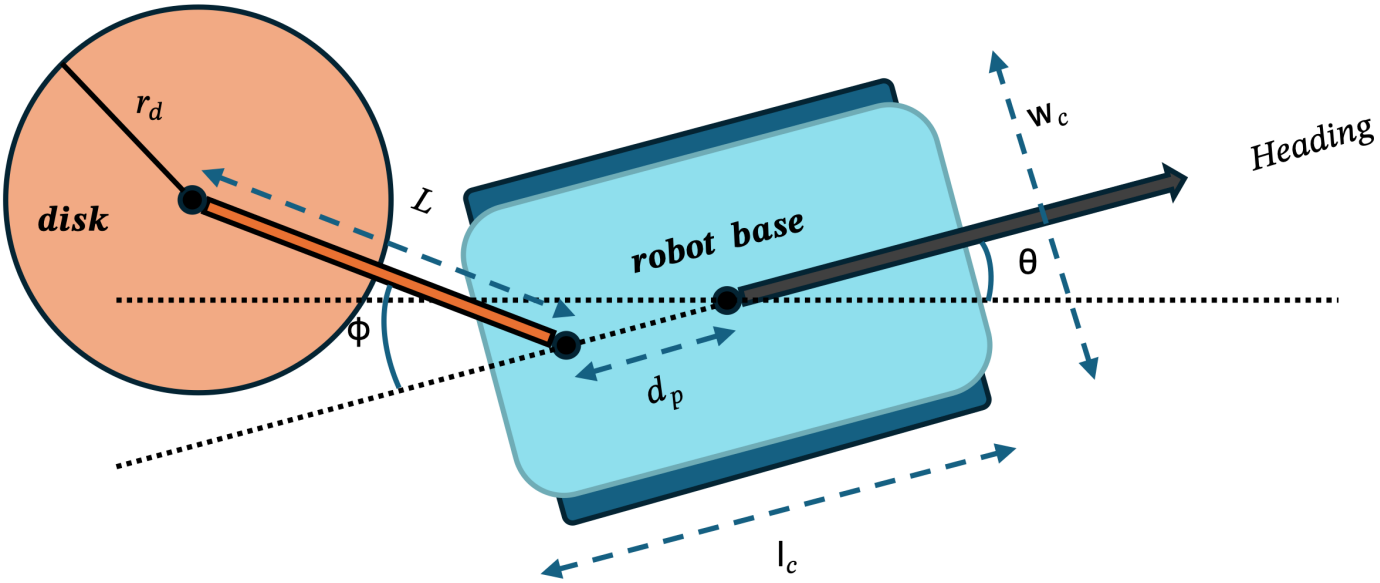
(d) complex, Fields2Cover
92.84%, 20.00 m^2 outside

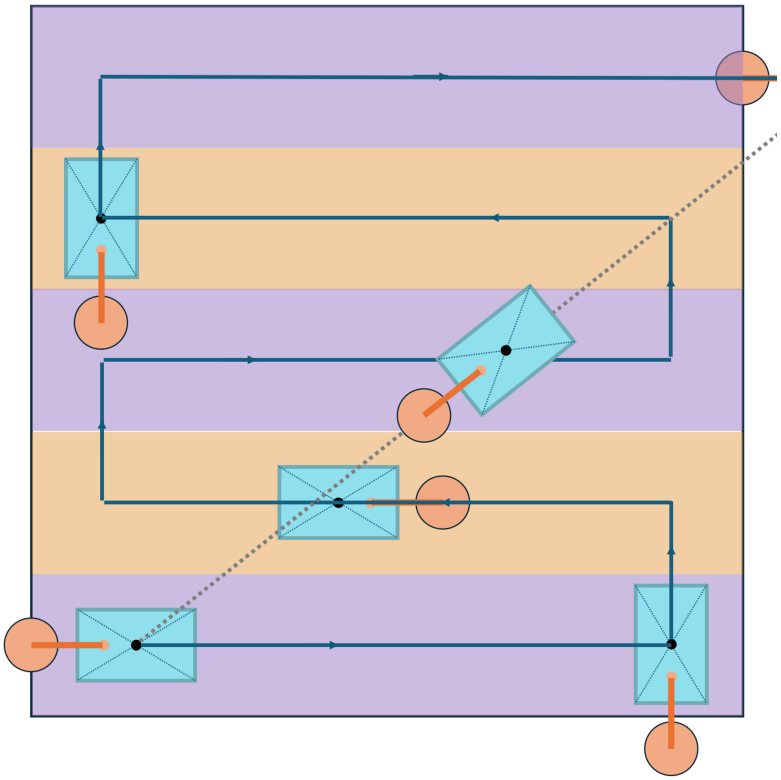


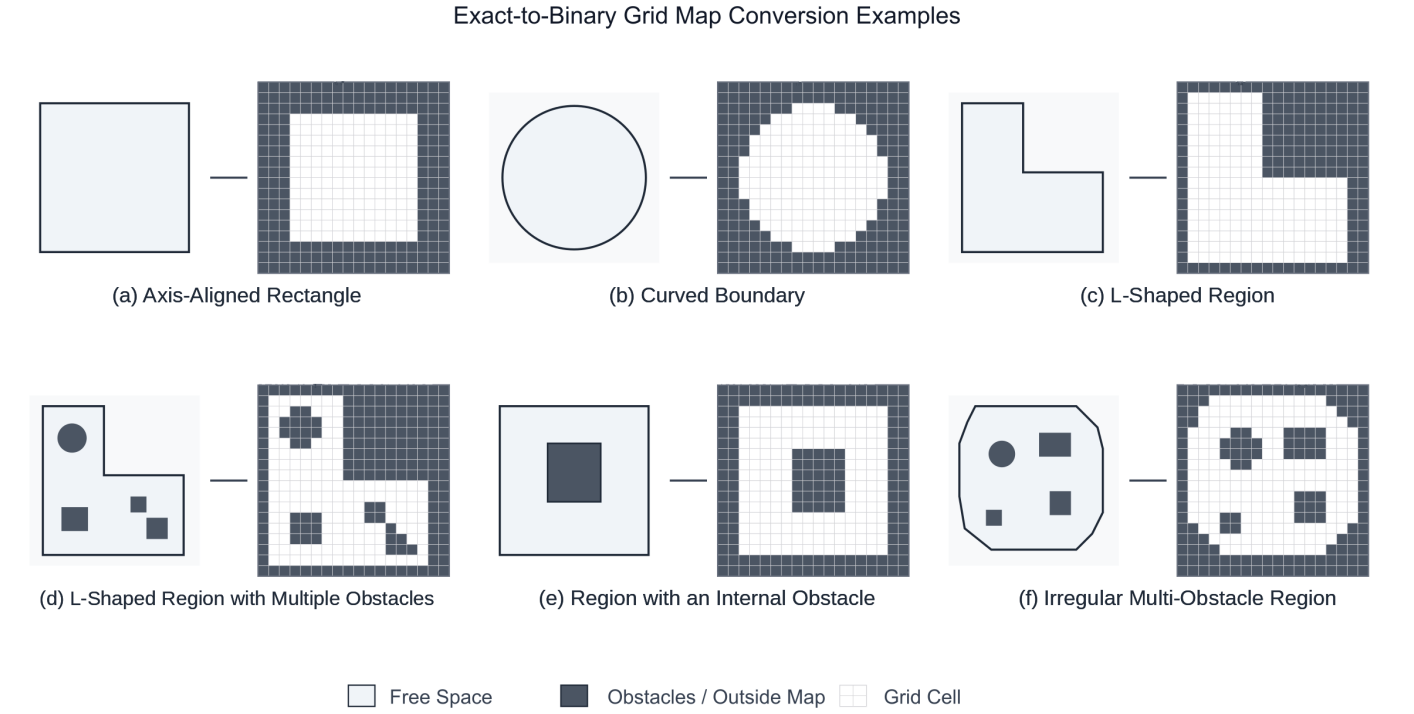




Baselines, unchanged


r_d
disk
L
robot base
w_c
Heading
θ
Φ
d_p
l_c




Exact-to-Binary Grid Map Conversion Examples
(a) Axis-Aligned Rectangle
(b) Curved Boundary
(c) L-Shaped Region
(d) L-Shaped Region with Multiple Obstacles
(e) Region with an Internal Obstacle
(f) Irregular Multi-Obstacle Region
Free Space
Obstacles / Outside Map
Grid Cell